Design and development of a glass-scintillator hadronic calorimeter for the CEPC reference detector

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Abstract

We present the design, technology development, and performance evaluation of a Glass-Scintillator Hadronic Calorimeter (GS-HCAL) proposed as the baseline hadronic calorimeter for the Circular Electron-Positron Collider (CEPC) reference detector. The CEPC is a proposed large circular e^+e^- collider designed to operate as a Higgs factory at $\sqrt{s} = 240$ GeV and as a high-luminosity Z-pole machine at $\sqrt{s} = 91.2$ GeV. The GS-HCAL employs high-density gadolinium fluoro-oxide (GFO) glass-scintillator cells of $40\times40\times10$ mm^3, each read out by a 3×3 mm^2 silicon photomultiplier (SiPM), arranged in a barrel and two endcaps totalling approximately 5.22 million channels. With a depth of $6\,\lambda_{\mathrm{I}}$ and a sampling fraction of approximately 30%, the design provides substantially higher active-material density than conventional sampling hadronic calorimeters. Key R&D results are reported: measured light yields of 60–100 p.e./MIP; radiation tolerance retaining more than 70% of the initial light yield at 100 Gy; and single-cell measurements with a ^{137}Cs source, cosmic-ray muons (63.6 ± 0.5 p.e./MIP), and 5 GeV electrons at KEK (157.6 ± 2.9 p.e./MIP). Full Geant4 simulation with a digitisation model yields a single-hadron energy resolution of $\sigma_E/E = 29.8\%/\sqrt{E} \oplus 6.5\%$ and an energy non-linearity within 2%. The Higgs-boson dijet mass resolution for $H \to gg$ achieved through particle-flow reconstruction is 3.88%, satisfying

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the CEPC target of 3–4%. Alternative technologies based on glass resistive-plate chambers (Semi-Digital HCAL, $65\%/\sqrt{E} \oplus 2.5\%$) and plastic scintillator tiles (Analog HCAL, $56\%/\sqrt{E} \oplus 2.5\%$) are also reviewed. A prototype beam-test programme spanning 2025–2027 will validate the full design.

Keywords: glass scintillator, hadronic calorimeter, CEPC, silicon photomultiplier, particle-flow algorithm, gadolinium fluoro-oxide, energy resolution, highly granular calorimeter

1. Introduction

The Circular Electron-Positron Collider (CEPC) [1] is a proposed large circular e^+e^- collider designed to operate as a Higgs factory and a high-luminosity Z-pole machine. Within the Particle-Flow Algorithm (PFA) paradigm [2, 3, 4], the Hadronic Calorimeter (HCAL) is responsible for measuring the dominant hadronic components of jets—comprising approximately 62–65% charged hadrons and 10% neutral hadrons, which collectively carry over 70% of the total jet energy. This capability is essential for precision measurements of dijet invariant masses, particularly in W/Z/Higgs-boson decays where the target boson mass resolution (BMR) of 3–4% [1] directly depends on the HCAL performance.

For charged hadrons the HCAL primarily provides shower-cluster shape information that is matched to tracks in the central tracker; the energy is then derived from the track momentum measurement. For neutral hadrons, which leave no tracks, the HCAL must independently provide both precise spatial reconstruction and accurate energy measurement. Good energy resolution also benefits the search for long-lived particles (LLPs) that may decay within the HCAL volume. These dual requirements demand an HCAL with exceptionally fine three-dimensional granularity combined with the best achievable energy resolution for neutral hadrons.

The Glass-Scintillator HCAL (GS-HCAL) [5, 6, 7] is proposed as the baseline design for the CEPC reference detector. It employs high-density gadolinium fluoro-oxide (GFO) glass-scintillator (GS) cells with a density of $6.0\,\mathrm{g/cm^3}$, offering crystal-like density at substantially reduced cost compared with bismuth germanate (BGO) crystals, while maintaining excellent moldability and scalability for mass production.

The geometric architecture comprises a barrel and two endcaps with a total depth of $6\,\lambda_\mathrm{I}$ in steel and GS. The barrel consists of 16 trapezoidal sectors arranged in a hexadecagonal cylinder with inner and outer radii of 2140 mm and

3455 mm, spanning 6460 mm along the beam axis. Each sector contains 48 longitudinal layers, progressively widening from 851 mm (front face) to 1375 mm (back face). The endcaps mirror this segmentation with 16 disk-shaped sectors per endcap, with inner and outer radii of 450 mm and 3455 mm. The design features 5.22 million identical GS cells of $40 \times 40 \times 10$ mm^3, read out by 3×3 mm^2 silicon photomultipliers (SiPMs) providing 60–100 p.e./MIP light output. The overall geometry is shown in Figure 1.

Geant4 simulations confirm that the design meets CEPC requirements, with a projected single-hadron energy resolution of $30\%/\sqrt{E} \oplus 6.5\%$ before calibration. A prototype beam-test programme spanning 2025–2027 will validate the design.

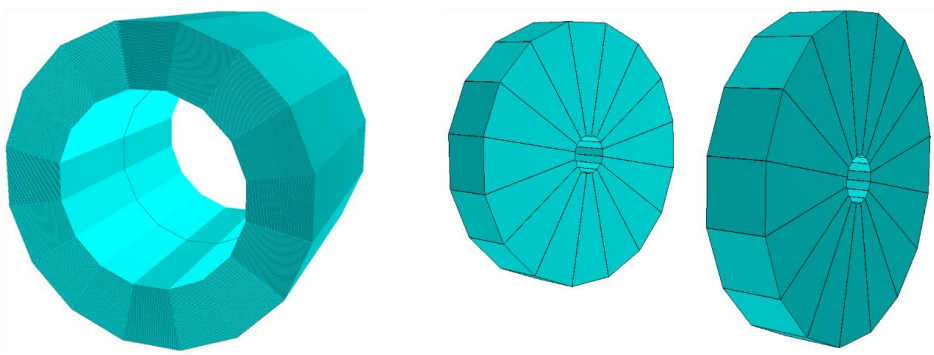


Figure 1: Geometric configuration of the GS-HCAL system. Left: barrel with 16 trapezoidal sectors forming a cylindrical structure (inner diameter 4280 mm, outer diameter 6910 mm). Right: endcap disks with matching sector segmentation.

The remainder of this paper is organised as follows. Section 2 describes the mechanical and geometrical design. Sections 3–6 present the GS material development, photon-detector characterisation, single-cell measurements, calibration strategy, readout electronics, and prototype plans. Section 7 reports the full simulation and physics performance. Section 8 reviews alternative HCAL technologies, and Section 9 summarises the results and outlines the R&D roadmap.

2. GS-HCAL design

This section details the engineering implementation of the GS-HCAL. The design translates the physics requirements into a modular mechanical structure with 48 longitudinal layers, achieving a sampling fraction of approximately 30% while minimising the proportion of non-sensitive material. Each 27.2 mm-thick layer

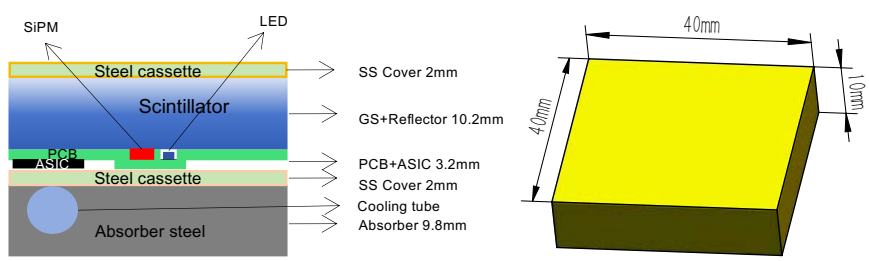
combines 9.8 mm steel absorbers with 10.2 mm GS active elements, maintaining a uniform cell granularity of $40 \times 40 \times 10$ mm^3 throughout the barrel and endcap.

2.1. Single-layer structure

The single-layer design of the GS-HCAL is illustrated in Figure 2. The calorimeter consists of alternating layers of GS cells (active medium) and steel plates (absorber), providing high density with fine segmentation for precise shower imaging.

Each layer has a total thickness of 27.2 mm, corresponding to approximately $0.125\,\lambda_{\mathrm{I}}$. From top to bottom the layer comprises: a 2 mm upper cover plate; the 10 mm GS active layer (40×40 mm^2 cells); a 3.2 mm printed-circuit-board (PCB) layer carrying application-specific integrated circuits (ASICs) for signal processing; a 2 mm bottom cover providing structural support; and the 9.8 mm steel absorber in which cooling pipes of 3 mm outer radius are embedded. One SiPM is positioned at the centre of each GS cell on a PCB that spans multiple cells.

The nuclear-interaction-length contributions of the steel absorber, GS, and PCB are $0.0805\,\lambda_{\mathrm{I}}$, $0.0425\,\lambda_{\mathrm{I}}$, and $0.0024\,\lambda_{\mathrm{I}}$ (ratio $1 : 0.53 : 0.03$), respectively, yielding a sampling fraction of approximately 31%.

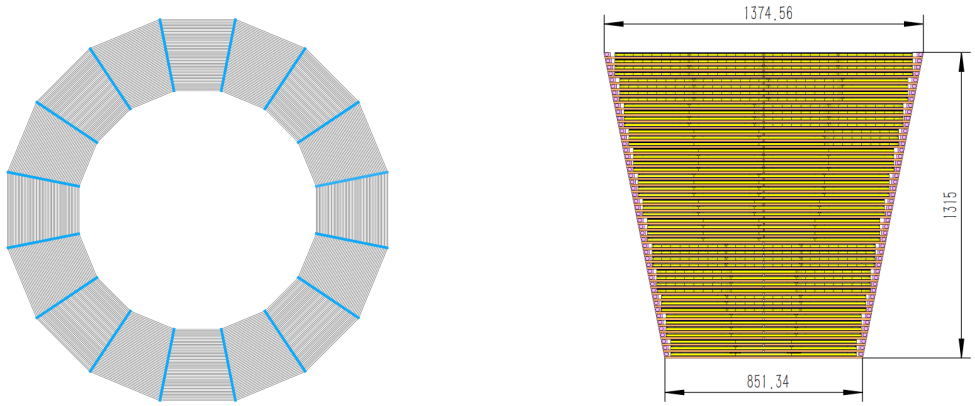


(a) (b)

Figure 2: (a) Single-layer structure of the GS-HCAL. (b) One GS cell of $40 \times 40 \times 10$ mm^3.

2.2. Barrel

The barrel is divided into 16 trapezoidal sectors (Figure 3), with inner and outer radii of 2140 mm and 3455 mm and a length of 6460 mm along the beam direction. Each sector has a front width of 851 mm and a back width of 1375 mm; its total thickness of 1315 mm corresponds to $6\,\lambda_{\mathrm{I}}$. The longitudinal segmentation into 48 layers follows the PFA optimisation studies performed with the Plastic-Scintillator HCAL (PS-HCAL) prototype [8]. The total barrel weight is 955 t.



(a) (b)

Figure 3: Dimensions of the GS-HCAL barrel. (a) Cross-sectional view showing 16 trapezoidal sectors (inner diameter 4280 mm, outer diameter 6910 mm). (b) Side view of a single trapezoidal sector (front width 851 mm, back width 1375 mm, depth 1315 mm).

The 16 radial sector boundaries introduce projective cracks, but effective coverage is maintained through compensating mechanisms. Hadronic showers predominantly initiate in the preceding BGO electromagnetic calorimeter (ECAL), which provides 1.2 λ_{I} of upstream material and ensures lateral spread that bridges sector boundaries. The solenoidal magnetic field provides additional muon curvature away from crack alignments.

Each layer consists of a steel absorber plate extending continuously over the full 6460 mm barrel length, upon which 30–40 modular detection boxes are precisely positioned. Each box contains a complete detection unit: GS cells, SiPMs, PCB, ASIC chips, cables, and cover plates. Taking the 48th (outermost) layer as an example (Figure 4), 40 boxes are arranged in 4 rows along ϕ with 10 boxes per row, each containing $8 \times 16 = 128$ cells and giving 5120 cells per layer. Each box is aligned with registration pins and secured with four corner bolts; a 2 mm gap between adjacent boxes accommodates thermal expansion.

The trapezoidal geometry requires adaptive box arrangements across the 48 layers. Three box widths (320 mm, 280 mm, and 240 mm) are combined to maintain full coverage while preserving the $40 \times 40 \times 10$ mm^3 cell granularity (Figure 5).

A water-cooling system manages the thermal load of the front-end ASICs (15 mW per channel). For the 48th layer the heat load is 76.8 W (5120 ch $\times$ 15 mW/ch). Each of the 48 layers in a sector incorporates four water-cooling pipes embedded in the steel absorber. Every eight consecutive layers share com-

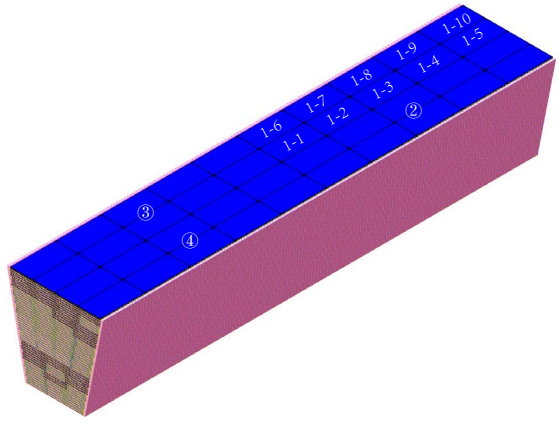


Figure 4: A barrel sector of the GS-HCAL, showing the arrangement of 40 boxes in the outermost (48th) layer. The four quadrants (1)–(4) each contain 10 boxes numbered n-1 to n-10.

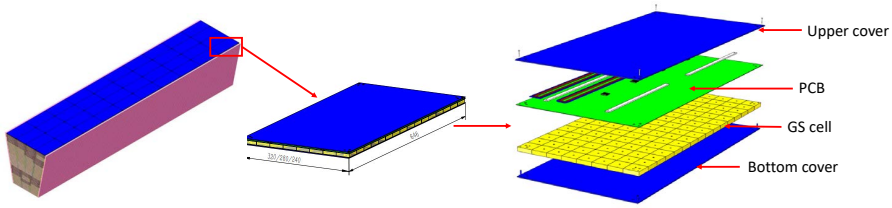
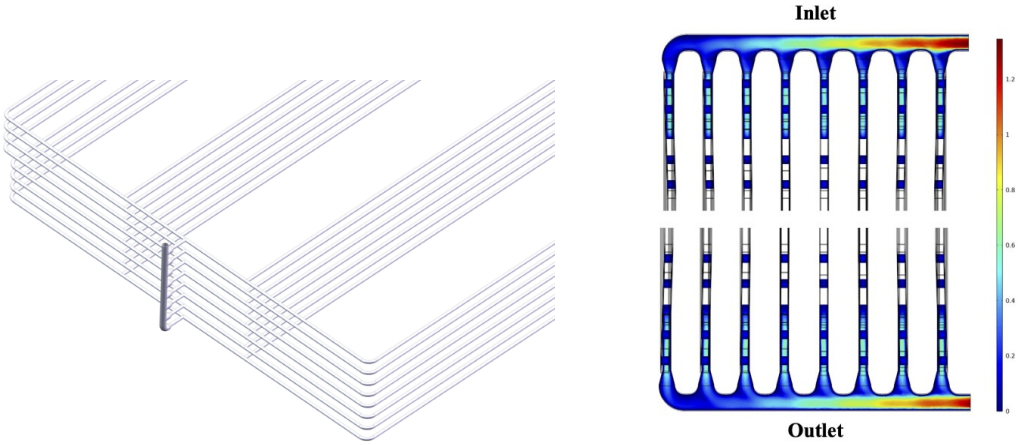


Figure 5: GS-HCAL barrel sector structure. Left (a): full sector composed of 48 layers. Centre (b): dimensions of an individual box (widths 240/280/320 mm, length 646 mm). Right (c): internal structure of a single box, including cover plates, GS cells, PCB with ASICs, cables, and cooling-water pipes.

mon input and output manifolds (Figure 6), resulting in six independent cooling loops per sector. Round pipes of 3 mm outer radius are used.



(a) (b)

Figure 6: Cooling-pipe routing for a barrel sector. (a) Schematic of the eight-in-one pipeline design. (b) Simulated coolant flow-velocity distribution (m/s), with inlet and outlet positions indicated.

The barrel component weights are summarised in Table 1.

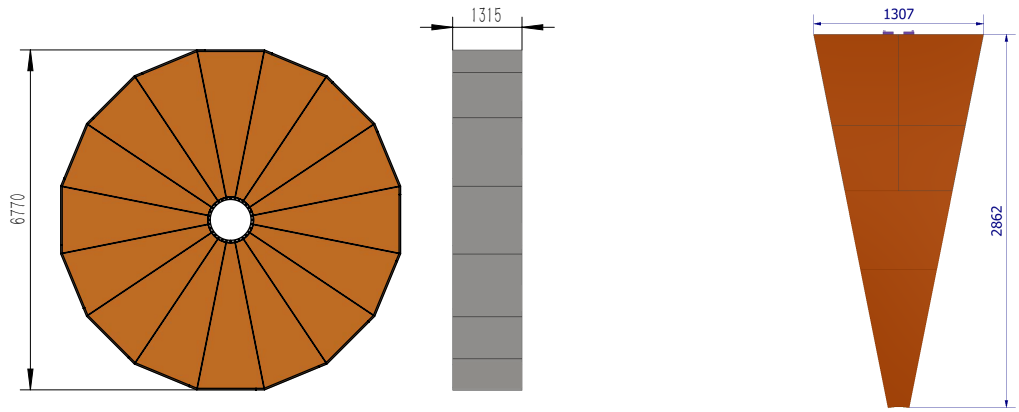
Table 1: Weight breakdown of the barrel GS-HCAL.

Component	**Weight/sector (kg)**	**Weight/barrel (t)**
Cover plate	10 287	164.6
PCB plate	390	6.2
Glass scintillator	19 277	308.4
Absorber layer	27 515	440.2
Support structure	1883	30.1
Auxiliaries	350	5.6
Total	59 702	955.2

2.3. Endcap

The two endcaps complement the barrel to provide full geometric coverage along the beam direction. Each endcap disk (Figure 7) matches the barrel design

with 16 sectors, inner and outer radii of 450 mm and 3455 mm, and a total thickness of 1315 mm divided among 48 layers. A single sector has a top width of 1307 mm and a radial height of 2862 mm, and is divided into 6 boxes along the radial direction. Each endcap weighs 362 t.



(a) (b)

Figure 7: Dimensions of the GS-HCAL endcaps. (a) Front and side views (full diameter 6670 mm, thickness 1315 mm). (b) Single sector; the complete endcap comprises 16 such sectors.

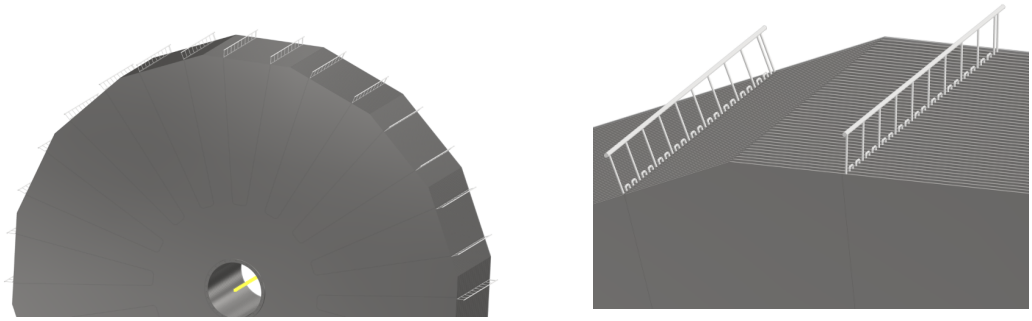
Construction begins with 1405 detector cells assembled into one sector layer; 16 sector layers form a complete disk-shaped layer, and 48 layers are stacked to complete the module. A 50 mm-thick outer structural plate provides rigidity and serves as the installation reference. Active layers and steel absorber plates are alternately installed horizontally and then rotated into the final vertical orientation.

The endcap cooling system provides 16 separate cooling regions per layer (Figure 8), each with one dedicated water channel. When all 48 layers are assembled the matching channels connect vertically, forming 48 parallel water circuits with 12 mm inner-diameter pipes.

2.4. Mechanical feasibility

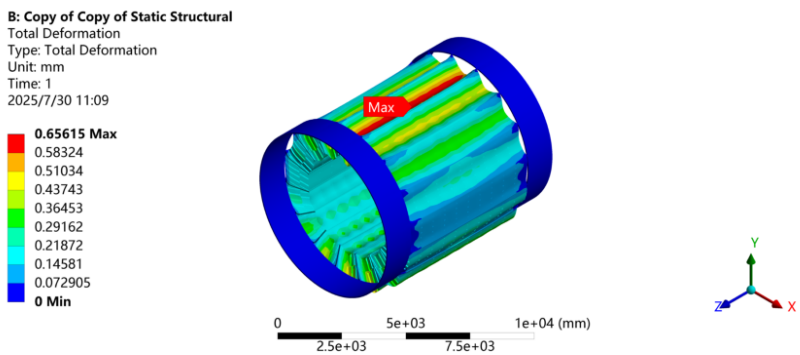
Structural finite-element analysis validates the mechanical feasibility of the GS-HCAL design. Particular attention was paid to the barrel's capacity to support the 135 t ECAL while preserving alignment precision. Figure 9 shows the deformation distribution with and without ECAL loading.

During assembly maximum deformations remain below 0.66 mm with localised stresses of 190 MPa, well below the yield strength of the structural materials. Under the full ECAL load, peak stresses reach 371 MPa at specific connection points;

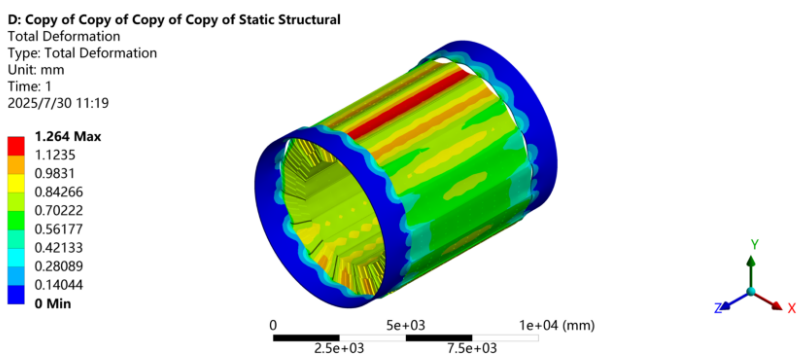


(a) (b)

Figure 8: Endcap cooling-pipe structure: (a) half of the endcap; (b) zoomed corner view.



(a)



(b)

Figure 9: Deformation distribution of the GS-HCAL (a) without and (b) with ECAL loading.

these critical areas employ titanium-alloy (TC4) components maintaining safety factors above 2.2, while the bulk of the structure uses standard stainless steel. Layer-to-layer deformation differences remain below 0.2 mm (within the 2.4 mm gasket accommodation range), and intra-layer deformations are constrained to less than 0.5 mm, producing only 20 MPa stress in the GS cells. The scintillator boxes withstand 0.5 mm bending with a safety factor exceeding three.

2.5. Channel count summary

Table 2 summarises the cell, box, layer, and sector counts. The complete detector contains approximately 5.22 million readout channels.

Table 2: Numbers of cells, boxes, layers, and sectors for the barrel and endcap.

Part	**Cells**	**Boxes**	**Layers**	**Sectors**
Barrel	3 212 800	27 840	48×16	16
Endcap ($\times 2$)	$1\,006\,080 \times 2$	3072×2	$48 \times 16 \times 2$	16×2

3. Glass-scintillator development

3.1. Historical context

The use of glass scintillators (GS) in calorimetry dates back to the 1980s, when the Crystal Clear Collaboration at CERN [9] pioneered investigations into high-Z glass matrices for electromagnetic calorimeters. The HED-1 GS prototype [9] demonstrated the material's potential but was limited by short radiation length and low light output; subsequent work on cerium-doped heavy-flint glass (HFG:Ce) revealed insufficient radiation hardness for high-luminosity environments. A persistent limitation of traditional GS was excessive self-absorption, which severely attenuated the light output in long-bar geometries and stifled numerous design proposals.

The maturation of SiPM technology and the adoption of PFA-based calorimetry in the 2010s fundamentally changed the picture [5, 6, 7]. Mass-produced SiPMs drastically reduced the cost of high-efficiency photodetection, while PFA granular segmentation eliminated the need for long light-transmission paths by employing compact, optically isolated cells. These innovations relaxed the light-yield requirements to a minimum of approximately 1000 ph/MeV, provided the SiPMs exhibit a photon detection efficiency (PDE) of $\geq 40\%$.

Post-2020 advances initiated by the Institute of High Energy Physics (IHEP) [10, 11, 12] focused on optimising the density–light-yield balance while ensuring manufacturing scalability. The GS-HCAL initiative, benefiting from the PS-HCAL design of the CALICE Collaboration [13, 8], employs GS cells with densities of approximately 6 g/cm^3 read out by SiPM arrays.

3.2. GFO glass: material properties

Over the past seven years the Glass-Scintillator Collaboration, led by IHEP and comprising universities, research institutes, and industrial partners, has systematically developed various glass compositions [10, 11, 12]. Figure 10 shows the light yield versus density for the GS samples investigated.

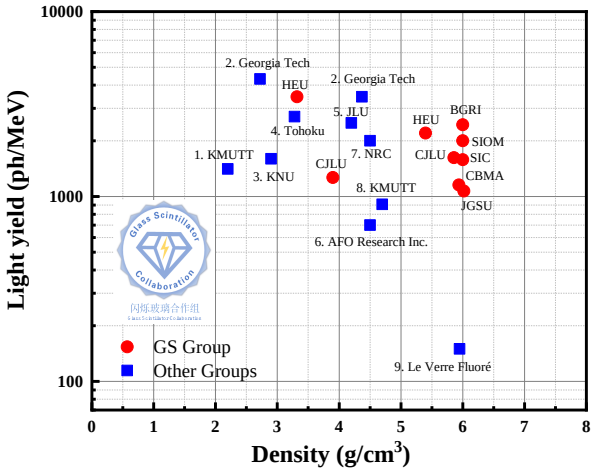


Figure 10: Light yield versus density for GS samples developed by the Glass-Scintillator Collaboration and other groups.

Gadolinium fluoro-oxide (GFO) glass was selected as the baseline material for its optimal balance between performance and manufacturability [14, 15, 16]. Table 3 compares its key parameters with those of BGO crystal [17] and DSB glass [18]. GFO achieves a density of 6.0 g/cm^3 with a radiation length of 1.59 cm and a nuclear interaction length of 24.2 cm. The composition features Ce^{3+} as the luminescent centre, chosen for its strong emission, fast decay, and efficient energy transfer from Gd^{3+} modifiers [19]. The borosilicate matrix (B_2O_3–SiO_2) provides excellent thermal and optical stability, with Al_2O_3 additives enhancing chemical

durability. Gadolinium serves as the heavy-element modifier, offering high density, low intrinsic radioactivity, and cost-effectiveness compared with lanthanum or lutetium alternatives.

Table 3: Key parameters of GFO glass ($5 \times 5 \times 5$ mm^3 sample) compared with BGO crystal [17] and DSB glass [18].

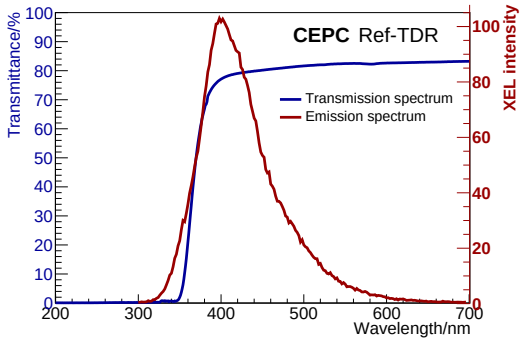
Parameter	**GFO**	**BGO**	**DSB**
Density (g/cm^3)	6.0	7.13	4.2
Melting point (°C)	1250	1050	1550
Radiation length (cm)	1.59	1.12	2.62
Molière radius (cm)	2.49	2.23	3.33
Nuclear interaction length (cm)	24.2	22.7	31.8
Z_{eff}	56.6	71.5	49.7
dE/dx (MeV/cm)	8.0	8.99	5.9
Emission peak (nm)	400	480	430
Refractive index	1.74	2.15	—
Light yield (ph/MeV)	~1500	7500	2500
Energy resolution (% at 662 keV)	~23	9.5	—
Scintillation decay time (ns)	~60, 500	60, 300	90, 400

3.3. Light yield

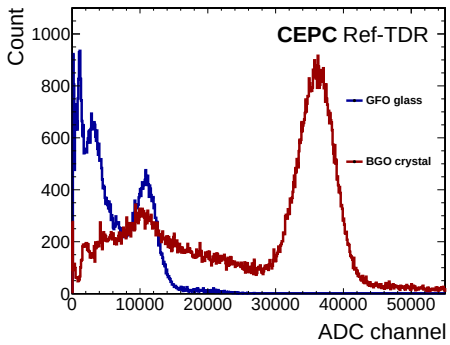
Figure 11 shows GFO glass cells of $40 \times 40 \times 10$ mm^3 under natural and ultraviolet light. Figure 12(a) presents the X-ray-excited luminescence (XEL) and optical-transmission spectra; the transmittance in the visible range exceeds 80%, with a self-absorption effect observed [20] near the 400 nm emission peak. Extensive γ-ray testing confirms that the GFO light yield consistently exceeds 1000 ph/MeV. When measured with an XP2020 (2 inch) photomultiplier tube (PMT), the detected photoelectron number reaches one-third that of BGO crystals of identical dimensions [21] (Figure 12b). Figure 12(c) shows the scintillation decay profile under γ-ray excitation: the slow-component decay time is controlled below 500 ns, which is fully compatible with the high granularity of the GS-HCAL design.

(a) (b)

Figure 11: GFO glass cells ($40 \times 40 \times 10$ mm^3) under (a) natural light and (b) ultraviolet illumination.



(a)



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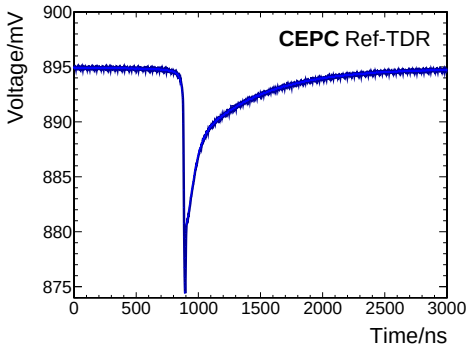


(c)

Figure 12: (a) Optical-transmission and XEL spectra of the GFO glass. (b) Energy spectra of GFO GS and BGO crystal of the same dimensions. (c) Scintillation decay curve of the GFO glass.

3.4. Light attenuation length

The light attenuation length (LAL, L_0) is determined through thickness-dependent transmittance measurements using [22]

$$L_0 = \frac{L_2 - L_1}{\ln(T_{L_1}/T_{L_2})}, \tag{1}$$

where L_1 and L_2 ($L_2 > L_1$) denote two sample thicknesses and T_{L_i} the corresponding transmittance values.

As shown in Figure 13, the best-performing GFO sample achieves $L_0 \approx 6$ cm at 400 nm (the GS emission peak). Across the visible spectrum above 400 nm the LAL remains stable between 6 cm and 6.8 cm.

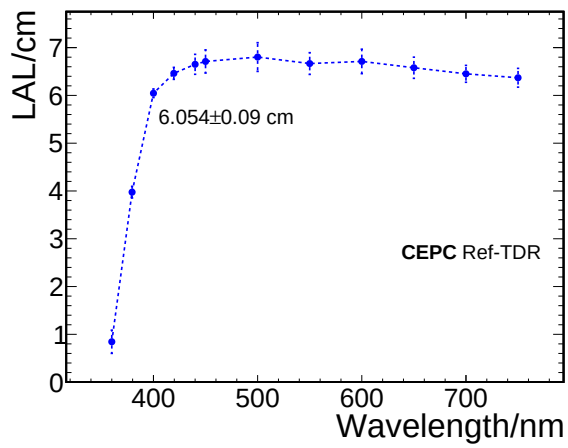


Figure 13: Attenuation length of GFO glass versus wavelength, measured using the transmittance method.

3.5. Radiation tolerance

Under the current CEPC operating plan the accumulated radiation dose reaches 40 Gy during the 10-year Higgs run (luminosity $\mathcal{L} = 8 \times 10^{34}$ cm^{-2}s^{-1}). For the Z-pole run ($\mathcal{L} = 192 \times 10^{34}$ cm^{-2}s^{-1}), one year of operation yields an estimated dose of approximately 96 Gy. As shown in Figure 14, the GFO light yield retains more than 70% of its initial value for doses below 100 Gy. This result is confirmed by proton-beam irradiation tests [23] at the APEP facility [24].

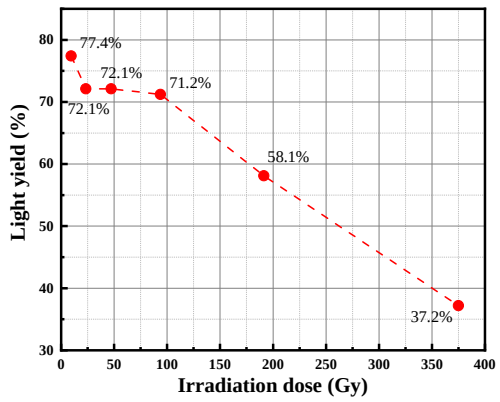


Figure 14: Relative light yield of GFO glass as a function of γ-irradiation dose. At 100 Gy the retention is 71.2%.

3.6. Quality control

Reproducible GS tile production with controlled quality is essential for the technology validation. Approximately 290 tiles of $40 \times 40 \times 10$ mm^3 have been produced to date; about 60% meet the target light yield exceeding 1000 ph/MeV. Tile-to-tile light-yield variations will be corrected through dedicated calibration, including a light-emitting-diode (LED) monitoring system and minimum-ionising-particle (MIP) inter-calibration using muon beams or cosmic rays.

A production run of approximately 10 000 tiles is scheduled for 2025–2026 to equip the full-size prototype. An automated batch-testing platform is being developed to measure tile properties rapidly and to enforce rigorous quality-control metrics: light-yield uniformity, attenuation-length verification, and dimensional tolerances.

4. Photon detector and optical simulation

4.1. SiPM selection and characterisation

Silicon photomultipliers (SiPMs) are selected for the GS-HCAL readout because of their compact form factor, high gain, low operating voltage, radiation hardness, and insensitivity to magnetic fields. SiPMs are already employed in the CMS High-Granularity Calorimeter (HGCAL) upgrade [25] and in CALICE highly-granular calorimeter prototypes [13].

Table 4 compares the parameters of representative commercial devices from Hamamatsu Photonics (HPK), the Novel Device Laboratory (NDL), and Rayquant (JBT). For the GS-HCAL application the primary selection criteria are high PDE at 400 nm, high gain, low dark count rate (DCR), and cost-effectiveness. The HPK S14160-3050 and NDL EQR20-11-3030 are identified as the most promising candidates for the prototype.

Table 4: Parameters of commercial SiPMs evaluated for the GS-HCAL.

Supplier	**HPK**		**NDL**	**JBT**
Type	S14160-3015PS	S14160-3050HS	EQR20-11-3030-S	JSP-TP3050-SMT
Pixel pitch (μm)	15	50	20	50
Number of pixels	39 984	3531	22 500	3364
Terminal capacitance (pF)	530	500	157.5	0.170
Breakdown voltage (V)	38 ± 3	38 ± 3	27.2 ± 1	24.6 ± 0.2
Recommended V_{op} (V)	V_{BD} + 3	V_{BD} + 2.7	V_{BD} + 5	V_{BD} + 2
Peak-sensitivity wavelength (nm)	450	450	420	420
Peak PDE (%)	32	50	47.8	35
PDE at 400 nm (%)	27	47	45	33
Gain	3.6×10^5	2.5×10^6	8.0×10^5	2.1×10^6
DCR (kHz/mm^2)	700–2100	10–100	150–450	120–270

The high DCR of SiPMs is a critical consideration for detection of weak signals. Figure 15 shows the DCR as a function of threshold level. At the baseline threshold of 0 p.e. the NDL EQR20 and HPK S14160-3050 devices exhibit DCRs of 255 kHz/mm^2 and 112 kHz/mm^2, respectively. Given a GS light output of 60 p.e./MIP, a 0.1 MIP threshold corresponds to approximately 6 p.e. At 5 p.e. the DCR drops to approximately 28 Hz/mm^2 (NDL) and 12 Hz/mm^2 (HPK), demonstrating the effectiveness of threshold optimisation for noise suppression.

4.2. Optical simulation of the GS cell

Geant4-based optical simulations [5] of a single GS cell coupled to a SiPM were performed to characterise the light-output performance and to identify optimisation pathways. The simulated geometry implements the baseline $40 \times 40 \times 10$ mm^3 cell with a single 3×3 mm^2 SiPM at its centre, the entire periphery wrapped in Teflon reflective film. The physics list `QGSP_BERT` is combined with a full set of optical processes: scintillation, Cherenkov radiation, Rayleigh scattering, bulk absorption, and boundary effects.

Figure 16(a) shows the light-output distribution for a simulated cell with 60 mm attenuation length, yielding 42.05 ± 0.01 p.e./MIP. Figure 16(b) confirms the expected increase in light-output most-probable value (MPV) with longer attenuation lengths.

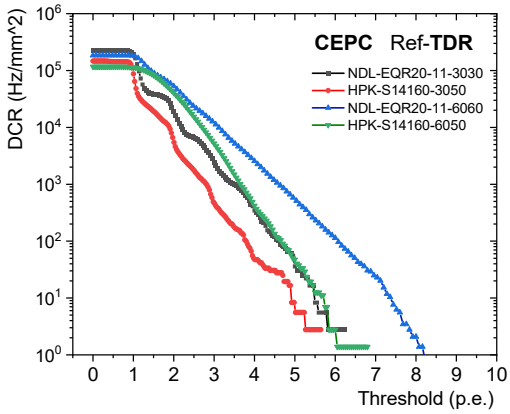
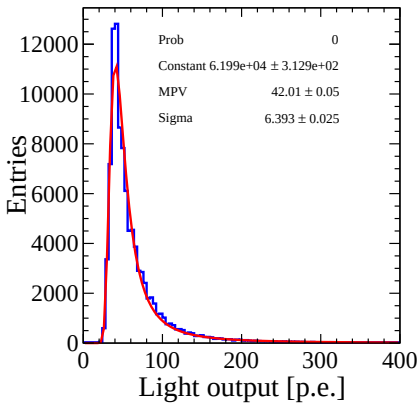
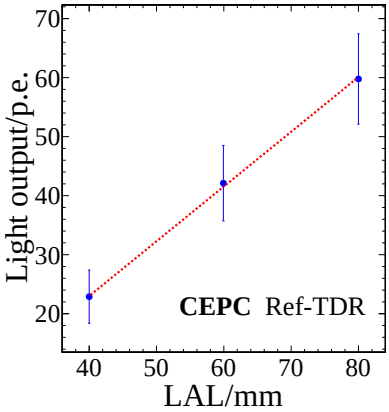


Figure 15: Dark count rate versus threshold level for several SiPM models.



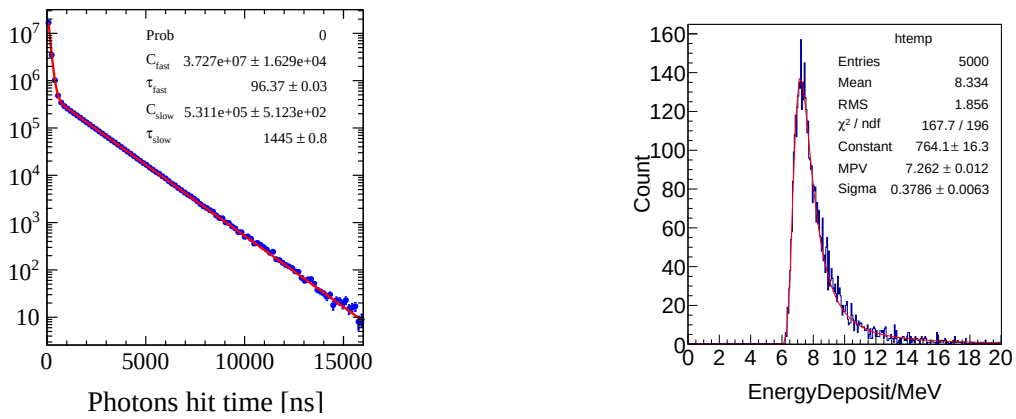
(a)



(b)

Figure 16: (a) Simulated light-output distribution of a GS cell with 60 mm attenuation length (42.05 ± 0.01 p.e./MIP). (b) Fitted light-output MPV versus attenuation length (40, 60, and 80 mm).

Validation studies (Figure 17) reveal simulated fast and slow scintillation-time components of 96 ns and 1445 ns, somewhat longer than the measured reference values (60 ns and 500 ns; Table 3), indicating that Geant4 fine-tuning with experimental GFO data is still required. The simulated deposited-energy distribution for 5 GeV muons yields an MPV of approximately 7.3 MeV per MIP, in good agreement with expectation.



(a) (b)

Figure 17: Simulation validation: (a) scintillation-time distribution (fast and slow components); (b) muon deposited energy in a 10 mm-thick GS cell.

5. Measurements and calibration

5.1. Single-cell measurements

The effective light output of $40 \times 40 \times 10$ mm^3 GS cells was evaluated in three independent measurement campaigns:

1. 662 keV γ-rays from a ^{137}Cs radioactive source;
2. cosmic-ray muons (expected energy deposition 7.7 MeV/MIP);
3. 5 GeV electron beams at KEK.

In each case the SiPM sensor was mounted at the cell centre on a readout board with a pre-amplifier. Comparative tests of enhanced-specular-reflector (ESR) and Teflon reflective films showed that Teflon provides approximately 20% higher effective reflectivity [26], attributed to its diffuse reflectivity and superior optical contact with the glass surface; all measurements therefore employ Teflon wrapping.

^{137}Cs measurement.. Using the HPK S14160-3050HS SiPM (spectrum-weighted PDE 44.5%), the MPV for 662 keV γ-rays is 12.9 p.e. Scaling to the expected 7.7 MeV MIP energy deposition yields an equivalent light output of approximately 150 p.e./MIP.

Cosmic-ray measurement.. A trigger formed by two plastic-scintillator panels selects through-going muons. With the HPK S13360-3050PE SiPM (spectrum-weighted PDE 32.5%) the measured light output is 63.6 ± 0.5 p.e./MIP (Figure 18b).

Electron beam test at KEK.. A Landau fit to the 5 GeV electron spectrum measured with the NDL EQR20-11-3030 SiPM (spectrum-weighted PDE 42.5%) yields an MPV of 157.6 ± 2.9 p.e./MIP (Figure 18c).

The factor-of-three increase in light output for 5 GeV electrons relative to cosmic muons arises from the fundamentally different energy deposition mechanisms. A cosmic muon deposits energy primarily through ionisation (~7.7 MeV/MIP), whereas a 5 GeV electron (well above the critical energy $E_c \sim 10$ MeV for GS) predominantly loses energy via bremsstrahlung, initiating an electromagnetic shower whose localised energy density greatly exceeds that of a minimum-ionising track.

The measured cosmic-ray light output of 64 p.e./MIP exceeds the simulated value of 42 p.e./MIP (Section 4.2). This discrepancy likely originates from uncalibrated Geant4 optical parameters for the newly developed GFO material. Improving the simulation accuracy remains a high priority.

5.2. Calibration strategy

5.2.1. Physics-process calibration

The global calibration of the GS-HCAL exploits MIP signals from prompt muons in $e^+e^- \to Z \to \mu^+\mu^-$ events. Figure 19 shows the kinematic distributions at $\sqrt{s} = 91.2$ GeV. At design luminosity this process occurs at a rate of 584 Hz. A conservative estimate yields 47 events per hour within a $\Delta\phi \times \Delta\cos\theta = 0.018 \times 0.018$ acceptance window corresponding to one GS cell at the central-barrel HCAL surface. Assuming bi-weekly Z-mode calibration runs (16 h/day), 4.71×10^8 di-muon events can be collected, achieving approximately 1% statistical precision.

The dominant Z-pole process $e^+e^- \to Z \to q\bar{q}$ provides resonant dijets for combined ECAL and HCAL calibration via the reconstructed invariant mass m_{jj} (Figure 20).

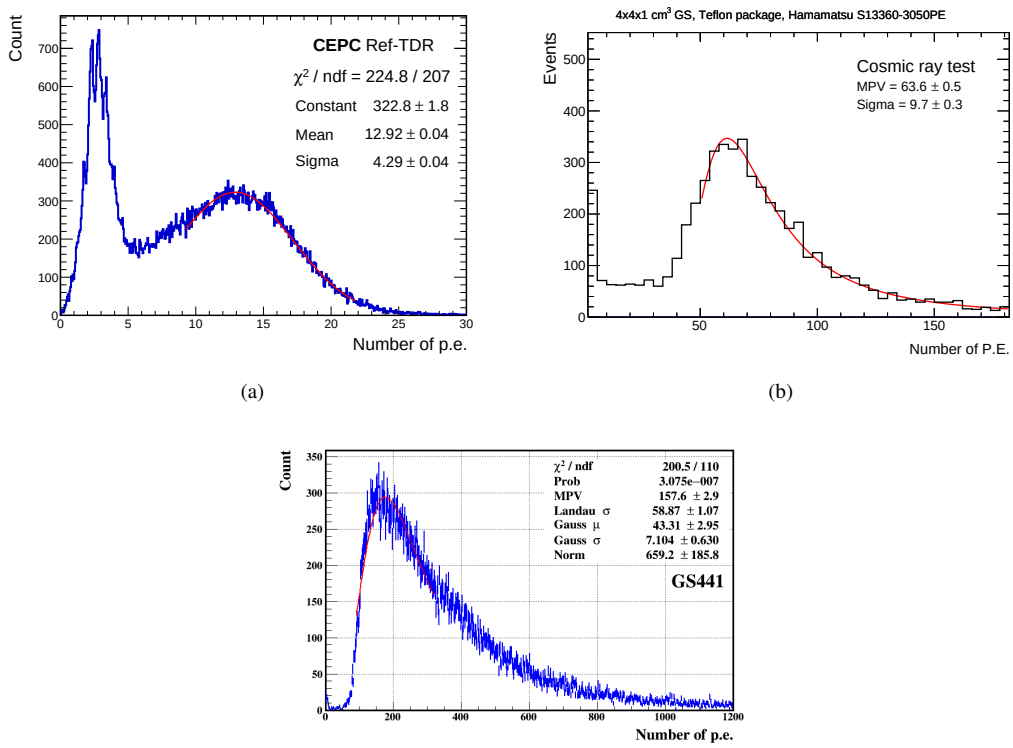


Figure 18: Single-cell measurement results ($40 \times 40 \times 10$ mm^3 GS with SiPM readout). (a) ADC spectrum with ^{137}Cs source (HPK S14160-3050HS). (b) Cosmic-ray test (HPK S13360-3050PE): 63.6 ± 0.5 p.e./MIP. (c) Response to 5 GeV electrons at KEK (NDL EQR20-11-3030): Landau-fit MPV 157.6 ± 2.9 p.e./MIP.

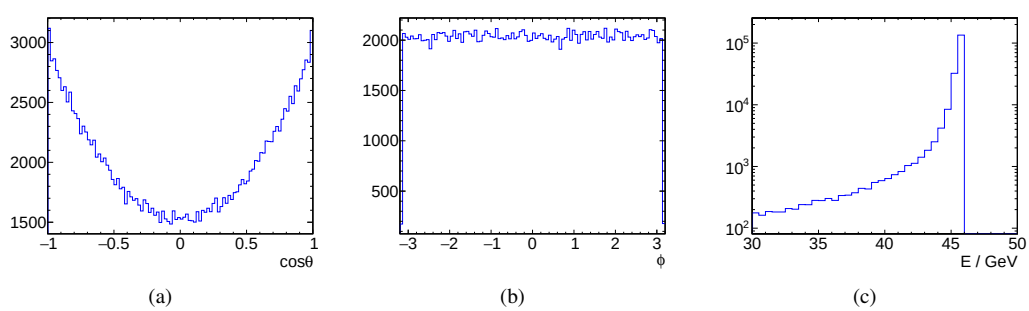


Figure 19: Generator-level distributions of (a) $\cos\theta$, (b) ϕ, and (c) energy of muons in $e^+e^- \to \mu^+\mu^-$ at $\sqrt{s} = 91.2$ GeV.

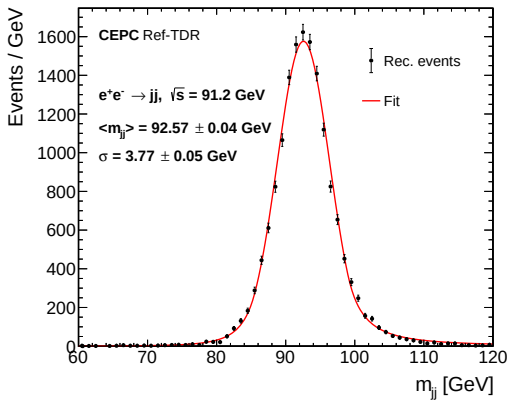


Figure 20: Distribution of dijet invariant mass from $Z \to q\bar{q}$ at $\sqrt{s} = 91.2\,\text{GeV}$.

Additional calibration using hadronic τ decays in $Z \to \tau^+\tau^-$ and isolated hadrons within jets provides complementary E/p monitoring of the HCAL response over time.

5.2.2. *Hardware calibration*

The hardware calibration system comprises two complementary approaches: (i) MIP inter-calibration using cosmic rays or muon beams, and (ii) an online LED-based monitoring system. The LED system (Figure 21) employs a centrally located LED with optical-fibre distribution to multiple cells, enabling continuous tracking of pedestal levels, gain characteristics, and response linearity.

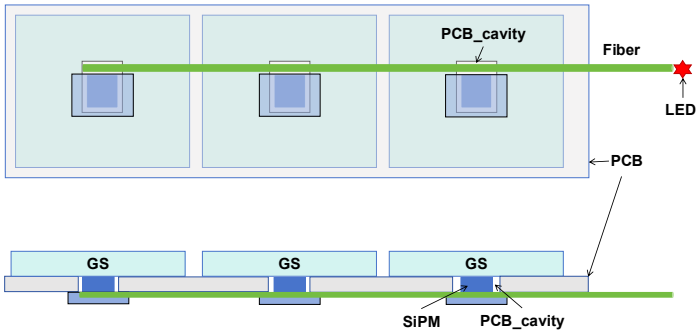


Figure 21: Schematic of the GS-HCAL LED calibration system showing optical-fibre connections between adjacent cells: top view (a) and side view (b).

6. Readout electronics and prototype

6.1. Readout electronics

The GS-HCAL requires front-end electronics (FEE) for more than five million SiPM channels. The readout chain comprises a stable, low-noise bias-voltage supply; a charge-sensitive pre-amplifier; a shaping stage optimised for signal-to-noise ratio; discriminators for photon counting; and analogue-to-digital converters (ADCs). Table 5 lists the key specifications.

Table 5: SiPM-driven requirements for the readout electronics.

Parameter	**Requirement**
Charge dynamic range	0.8 pC–800 pC (0.1–100 MIP)
Charge resolution	10% of 1.0 MIP
SiPM capacitance	≤ 100 pF
SiPM gain	$\geq 5 \times 10^5$
Average event rate/channel	2 kHz
Maximum event rate/channel	50 kHz
Typical signal amplitude	2–3 mV/p.e.

An interim FEE solution [27] was developed for SiPM characterisation and prototype testing, as no commercial option met the requirements. The pre-amplifier (Figure 22) achieves a gain of ± 20 V/V, bandwidth of 400 MHz, and baseline noise of $300\,\mu\mathrm{V}_{\mathrm{RMS}}$ with $50\,\Omega$ input impedance. Figure 22(e) shows a typical SiPM waveform, and Figure 22(f) the ADC spectrum of dark noise with well-separated photoelectron peaks.

6.2. Prototype design and beam-test programme

A GS-HCAL prototype is under construction following the single-layer design of Section 2.1. The prototype is a steel-scintillator sandwich of $52.0 \times 52.0 \times 130.6$ cm^3 (≈ 0.6 m^3), containing 8112 GS cells in 48 layers ($\approx 6\,\lambda_{\mathrm{I}}$). Each layer has a 13×13 array of $40 \times 40 \times 10$ mm^3 cells, each coupled to a 3×3 mm^2 SiPM. The steel absorber plates are S304 stainless steel (9.8 mm thick).

Geant4 simulations with 80 GeV π^- beams (Figure 23) show that the $13 \times 13 \times 48$ configuration achieves 95% energy containment. Figure 24 presents the projected energy resolution and linearity. The prototype is expected to achieve a single-hadron energy resolution of approximately $45\%/\sqrt{E} \oplus 8\%$, degraded relative to the full detector owing to the limited lateral size; this performance is

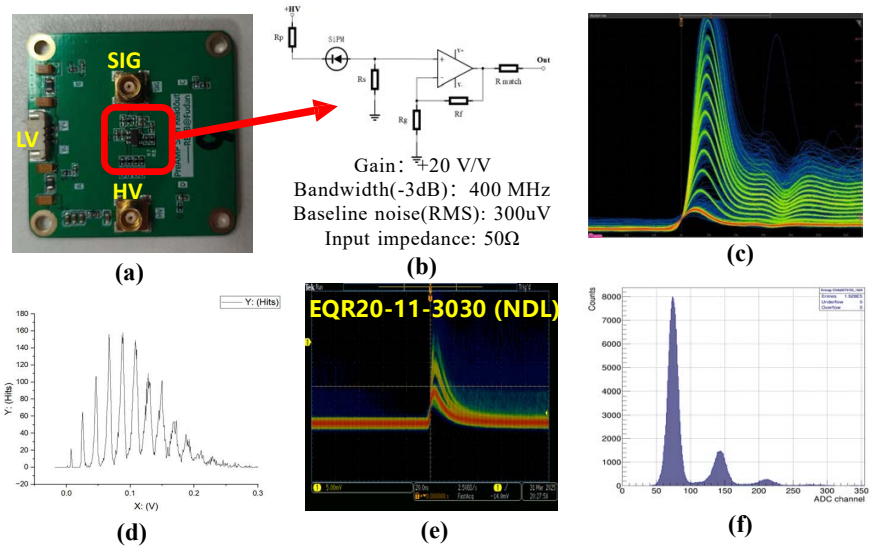


Figure 22: Pre-amplifier for the GS-HCAL SiPM readout. (a) Single-channel PCB coupled to a GS cell. (b) Schematic diagram. (c, d) Performance characteristics. (e) Waveform of a 3×3 mm^2 NDL SiPM (EQR20-11-3030D-S). (f) ADC spectrum of SiPM dark noise.

nonetheless sufficient to validate the key GS technology and reconstruction algorithms.

The construction and testing programme proceeds in phases:

- A $3 \times 3 \times 7$ mini-prototype is under construction for initial beam tests at CERN in October 2025.
- The full 8112-channel prototype is scheduled for completion by end-2026, with cosmic-ray tests at IHEP and beam tests preferably in 2027.
- The beam-test platform provides ±25 cm positioning accuracy (< 1 mm precision), ±30° rotation, and 5 t capacity.
- Alternative test sites include the China Spallation Neutron Source (CSNS, 1.6 GeV proton beam) and the facility at the University of Tokyo.

Key subsystems include: blue-LED calibration for SiPM-gain monitoring; a new ASIC with 15 mW/channel power consumption; TiO_2 and Teflon reflective coating; and a water-based cooling system with precisely regulated flow rate and temperature.

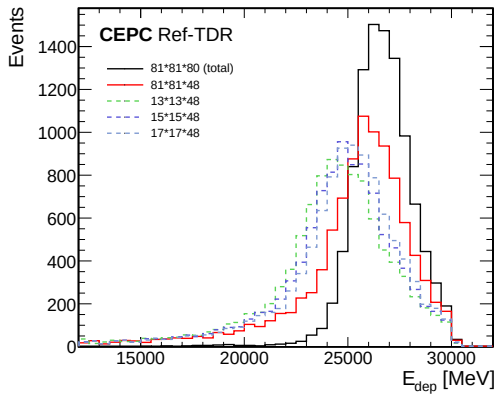
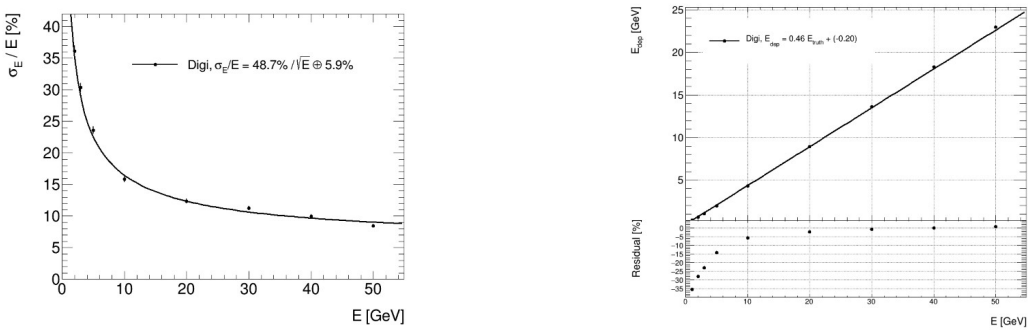


Figure 23: Simulated deposited energy in prototypes of different lateral sizes for 80 GeV π^-; the $13 \times 13 \times 48$ configuration achieves ≈95% containment.



(a) (b)

Figure 24: Simulated (a) energy resolution and (b) energy linearity of the GS-HCAL prototype.

7. Simulation and performance

7.1. Simulation setup

The intrinsic energy linearity and resolution of the GS-HCAL are evaluated using Geant4 simulation within the CEPCSW framework [1], with all inner sub-detectors removed. A digitisation model incorporates the following effects:

- **Birks' constant:** a preliminary value $k_B = 0.01\,\mathrm{cm/MeV}$ is used; a dedicated measurement with heavy-ion beams is being planned.
- **Light yield:** based on GS measurements the intrinsic MIP light yield is 1500 ph/MeV, giving a detected effective light output of ≥ 60 p.e./MIP.
- **Attenuation length:** 6 cm (current measured value).
- **Readout threshold:** 0.1 MIP, determined by scanning its impact on the energy resolution.
- **SiPM response** and electronic-system characteristics.

7.2. Single-hadron energy resolution

The energy resolution for single hadrons is parametrised as

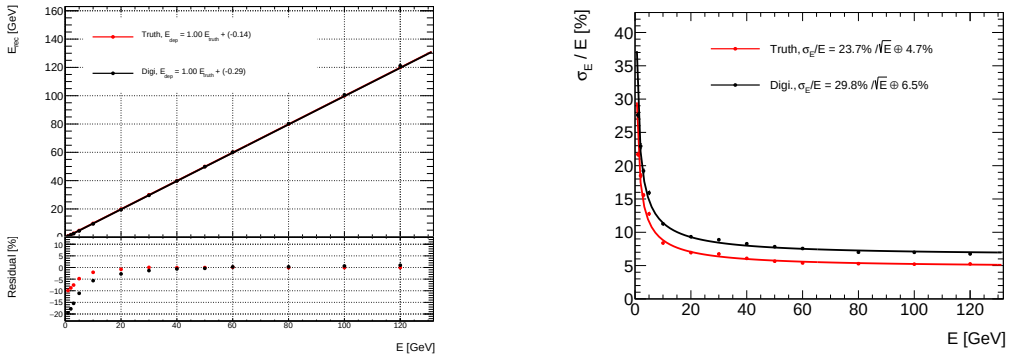
$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus b, \tag{2}$$

where a denotes the stochastic term and b the constant term. With the full digitisation model the GS-HCAL yields

$$\frac{\sigma_E}{E} = \frac{29.8\%}{\sqrt{E\,[\mathrm{GeV}]}} \oplus 6.5\%, \tag{3}$$

with energy non-linearity within 2% (Figure 25). The default hadronic-physics list is `QGSP_BERT`. The stochastic term substantially outperforms conventional HCALs, which typically achieve $\sim 60\%/\sqrt{E} \oplus 3\%$. The comparatively large constant term is attributed to longitudinal leakage in the $6\,\lambda_\mathrm{I}$ design and is expected to be further reduced by software compensation and PFA techniques [28, 29, 30, 31].

In the CEPC collision environment (Figure 26), more than 98% of hadronic final-state objects have energies below 60 GeV, a regime in which the GS-HCAL achieves optimal resolution.



(a) (b)

Figure 25: (a) Energy linearity and (b) energy resolution of the GS-HCAL from the full digitisation model.

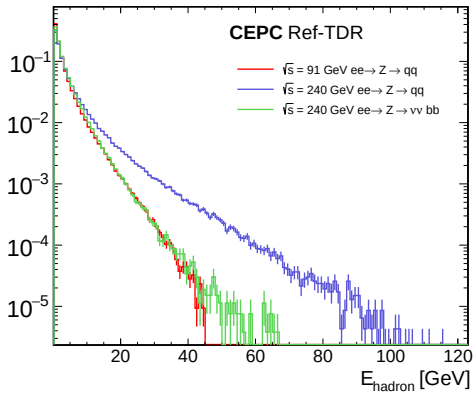
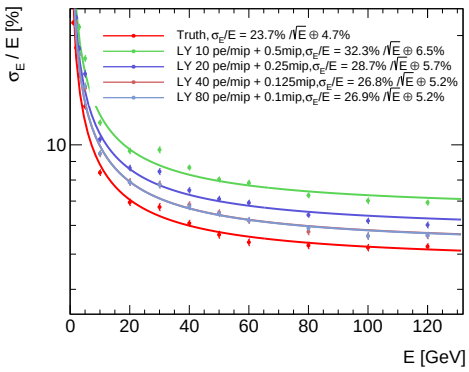
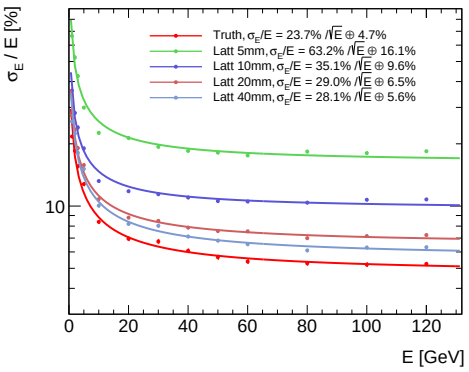


Figure 26: Energy distribution of hadronic objects in the most energetic CEPC e^+e^- processes.



(a)



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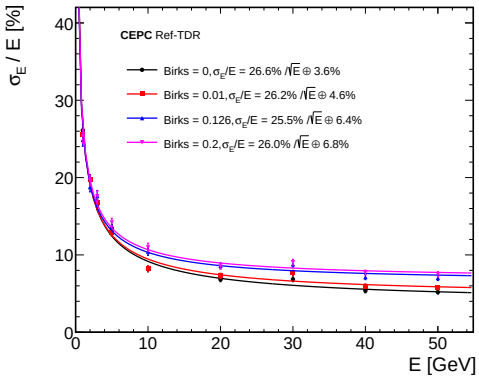
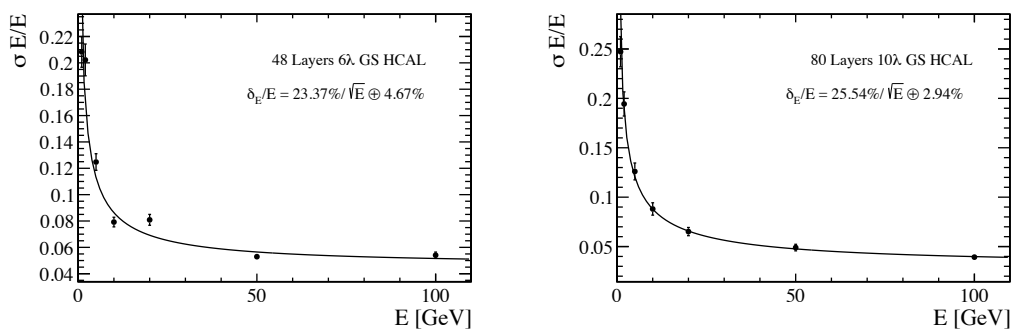


(c)

Figure 27: Simulated energy resolution as a function of (a) light output and readout threshold, (b) GS attenuation length, and (c) Birks' constant.

A systematic parameter scan (Figure 27) quantifies the sensitivity of the energy resolution to (a) light output and threshold, (b) GS attenuation length, and (c) Birks' constant, confirming that these parameters are most influential in the low-energy regime.

To confirm that longitudinal leakage dominates the constant term, the GS-HCAL depth was increased from 48 layers ($6\,\lambda_{\mathrm{I}}$) to 80 layers ($10\,\lambda_{\mathrm{I}}$); the constant term dropped from 4.7% to 2.9% (Figure 28). With the full detector the ECAL ($\approx 1.2\,\lambda_{\mathrm{I}}$) causes approximately 70% of hadrons to initiate showers upstream, which is expected to reduce leakage significantly. Requiring the first hadronic interaction to occur within the first two GS-HCAL layers ($0.25\,\lambda_{\mathrm{I}}$) already reduces the constant term to approximately 3%, consistent with PS-HCAL prototype measurements [8, 31].



(a) (b)

Figure 28: Comparison of energy-resolution distributions for two GS-HCAL depths: (a) $6\,\lambda_{\mathrm{I}}$ (48 layers), showing a pronounced high-energy tail; (b) $10\,\lambda_{\mathrm{I}}$ (80 layers), with the constant term reduced from 4.7% to 2.9%.

7.3. Electron–hadron response

The calorimeter response to electrons and various hadrons was studied using full Geant4 simulations of the standalone GS-HCAL (no upstream ECAL). The true energy E_{true} is fitted against the raw calorimeter signal S_{HCAL} using a linear model $E_{\mathrm{true}} = C \times S_{\mathrm{HCAL}}$, extracting a calibration constant C for each particle species. The ratio C_{hadron}/C_e, summarised in Table 6, quantifies the degree of non-compensation: a value exceeding unity indicates that the hadron of a given energy produces a smaller signal than an electron of the same energy. The measured ratios of 1.09–1.14 are consistent with expectations for a non-compensating sampling calorimeter [32].

It is important to note that C_{hadron}/C_e, as an empirically measured calibration ratio, is distinct from the intrinsic compensation ratio e/h defined in the context of hadronic shower physics. The intrinsic e/h is the true figure of merit: a value of unity signifies perfect compensation and eliminates the degrading effect of fluctuations in the electromagnetic shower fraction f_{em} on the energy resolution [32]. Future work will extract the e/h value through detailed Geant4 simulations that separate the electromagnetic and hadronic components of hadronic showers. By analysing the origin of energy deposits, the average electromagnetic fraction $\langle f_{\text{em}} \rangle$ and its event-by-event fluctuations can be determined. A precise calibration exploiting the intrinsic e/h parameter—via software compensation or energy-weighting algorithms—is expected to reduce the constant term in the energy resolution below 3%.

Table 6: Calibration slopes C and electron–hadron response ratios C_{hadron}/C_e for the GS-HCAL.

Particle	C (slope)	C_{hadron}/C_e
Electron	2.94 ± 0.01	—
π^-	3.20 ± 0.05	1.09 ± 0.02
K^+	3.28 ± 0.05	1.12 ± 0.02
Neutron	3.35 ± 0.07	1.14 ± 0.02
Proton	3.36 ± 0.07	1.14 ± 0.02

7.4. Physics performance: Higgs-boson mass resolution

The detector-level physics performance is evaluated through PFA reconstruction of the benchmark process $H \to gg$. Event generation uses WHIZARD v1.9.5 [33, 34] with next-to-leading-order matrix elements; the full detector response is modelled with Geant4 v11.2 within the CEPCSW framework. Figure 29 shows the reconstructed dijet invariant mass. The fitted Higgs-boson mass is 126.32±0.04 GeV/c^2 with $\sigma(m_{jj}) = 4.90 \pm 0.04$ GeV/c^2, corresponding to a boson mass resolution (BMR) of **3.88%**, satisfying the CEPC target of 3–4%.

8. Alternative HCAL technologies

Extensive R&D within the CALICE Collaboration has produced well-validated alternative technologies for PFA-based hadronic calorimetry [8, 35, 36]. These include the Digital HCAL (DHCAL) [37, 38] and Semi-Digital HCAL (SDHCAL) [35, 36] using gaseous detectors, as well as the Analog HCAL (AHCAL) based on

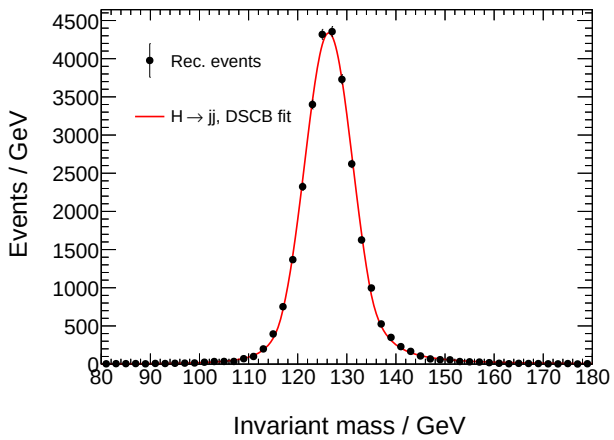


Figure 29: Reconstructed dijet invariant mass for $H \to gg$. Fitted mass: 126.32 ± 0.04 GeV/c^2; $\sigma(m_{jj})$ = 4.90 ± 0.04 GeV/c^2; BMR = 3.88%.

plastic scintillator tiles [8]. The dual-readout concept [39, 40, 41, 42, 43] has also been investigated, though the CEPC baseline remains firmly within the PFA paradigm. The two most advanced PFA-compatible alternatives are summarised below.

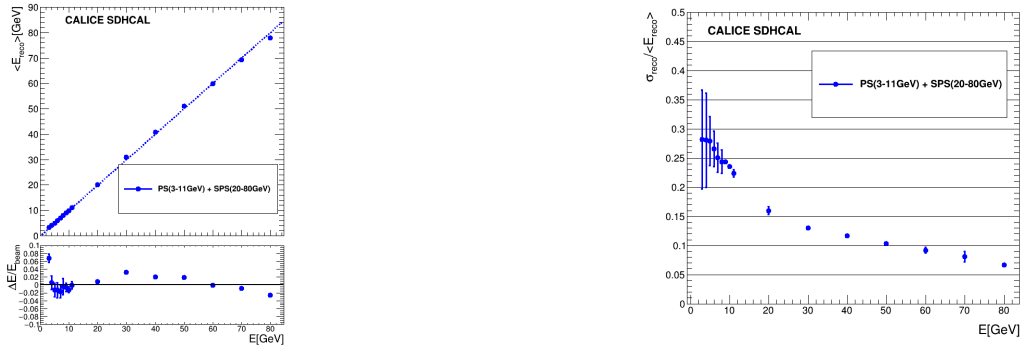
8.1. RPC-based Semi-Digital HCAL

The SDHCAL [35, 44] employs glass resistive-plate chambers (GRPCs) with 1×1 cm^2 pick-up pads and HARDROC2 ASICs [45] providing 2-bit semi-digital readout. The prototype consists of 48 active layers inserted into a self-supporting structure of 49 stainless-steel plates (15 mm each), totalling $\approx 6\,\lambda_{\mathrm{I}}$. Multi-threshold readout mitigates saturation effects at energies above $\sim$30 GeV [44].

Beam tests at CERN PS and SPS with π^- beams from 3 to 80 GeV [44, 46] yield an energy linearity within $\approx$4% and a resolution of $65\%/\sqrt{E} \oplus 2.5\%$ (Figure 30).

8.2. Plastic-scintillator Analog HCAL

The AHCAL [8, 47, 31] uses injection-moulded plastic-scintillator tiles ($40 \times 40 \times 10$ mm^3, later 3.5 mm thick) wrapped in ESR film and read out by SiPMs. It was the baseline HCAL design in the CEPC Conceptual Design Report (CDR), and remains a mature and well-validated fallback technology. Extensive simulation studies have addressed optimal absorber thickness, sampling-layer count, and cell size [47, 31]. A technological prototype of $72 \times 72 \times 120$ cm^3 (40 layers,



(a) (b)

Figure 30: SDHCAL prototype beam-test results [46]: (a) energy linearity; (b) energy resolution.

4.7 λ_I, 12 960 channels) was developed by USTC, SJTU, and IHEP, and tested at CERN SPS and PS. Approximately 15 000 plastic tiles were characterised with a ^{90}Sr source, yielding a mean light output of ≈12.9 p.e./MIP with 91.6% of tiles within a 10% uniformity window [48].

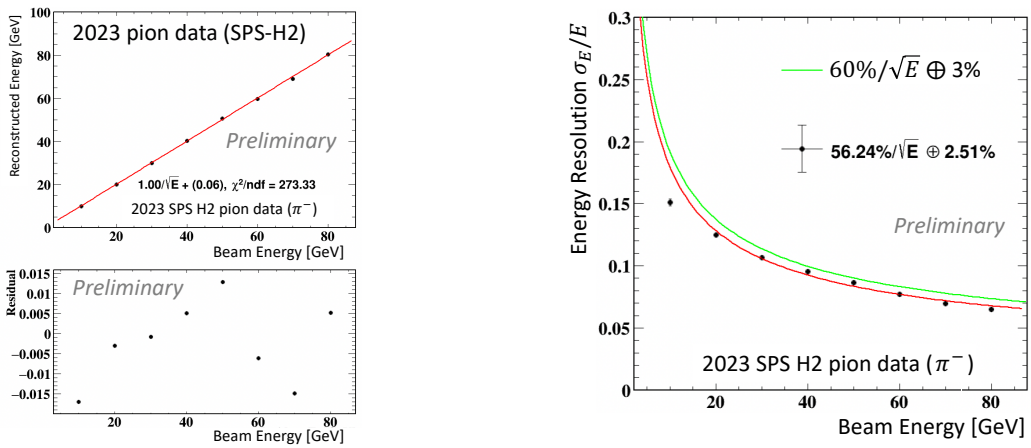
The readout uses SPIROC2E ASICs [49, 50, 51] with 36 channels each and dual-gain preamplifiers. Three beam-test campaigns at CERN collected approximately 65 million events, and advanced machine-learning methods were applied for particle identification [52]. The measured energy linearity is better than 1.5% and the energy resolution is $56.2\%/\sqrt{E} \oplus 2.5\%$ [31] (Figure 31). Software-compensation algorithms exploiting the high granularity can improve the hadronic energy resolution by up to 30% [53, 28].

9. Summary and outlook

The GS-HCAL has been presented as the baseline hadronic calorimeter for the CEPC reference detector, employing high-density GFO glass-scintillator cells read out by SiPMs in a barrel and two endcaps totalling approximately 5.22 million channels.

The principal R&D achievements reported in this paper are:

- Development of GFO glass with a light yield exceeding 1000 ph/MeV, an attenuation length of ≈6 cm, and radiation tolerance retaining more than 70% of the initial light yield at 100 Gy.
- Single-cell measurements demonstrating 63.6 ± 0.5 p.e./MIP with cosmic muons and 157.6 ± 2.9 p.e./MIP with 5 GeV electrons at KEK.



(a) (b)

Figure 31: AHCAL prototype performance with high-energy π^- beams [31]: (a) energy linearity (better than 1.5%); (b) energy resolution ($56.2\%/\sqrt{E} \oplus 2.5\%$, red curve).

- A custom pre-amplifier FEE with 400 MHz bandwidth and 300 μV_{RMS} noise.
- Geant4 simulation yielding $\sigma_E/E = 29.8\%/\sqrt{E} \oplus 6.5\%$ and a Higgs BMR of 3.88% for $H \to gg$.
- A validated mechanical design with deformation below 0.66 mm under assembly loads and adequate margin when supporting the 135 t ECAL.

The constant term of ≈5–6%, while sufficient for the 3–4% BMR target, can be further reduced below 3% through: (i) improved calibration exploiting the intrinsic e/h ratio extracted from shower-component decomposition; (ii) energy-weighting and software-compensation algorithms; (iii) position-dependent light-collection efficiency corrections; and (iv) machine-learning-based shower reconstruction [28, 29, 30, 31].

The GS-HCAL development follows a phased timeline:

- **2024–2025:** finalise the $40 \times 40 \times 10$ mm^3 tile design and begin mass production of ~10 000 prototype tiles.
- **2026:** complete the 0.6 m^3 prototype (8112 tiles) and begin beam tests at CSNS or CERN.
- **2027:** integrate a full-scale module and validate performance with cosmic-ray and test-beam data.

Ongoing and planned R&D includes: further optimisation of the GS light output and attenuation length; evaluation of emerging SiPM devices; development of a custom low-power (15 mW/channel) ASIC for the final readout system; and integration of the HCAL electronics into the CEPC-wide data-acquisition architecture.

The SDHCAL and AHCAL alternatives, with beam-test-confirmed energy resolutions of $65\%/\sqrt{E} \oplus 2.5\%$ and $56\%/\sqrt{E} \oplus 2.5\%$, respectively, provide important benchmarks and contingency pathways. The results presented in this paper establish a strong foundation for realising a high-performance, cost-effective hadronic calorimeter for the CEPC.

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